

Question number	Answer	Additional guidance	Mark
2(a)	An answer that includes two of the following points: - cells that are {dividing / multiplying} out of control (1) - caused by a {mutation / change in (DNA) base sequence} (1) - cells can break off (from some cancers) / and spread {through the body / to other tissues} (1)	ACCEPT rapid cell division / mitosis no Hayflick's limit IGNORE growth ACCEPT tumour suppressor gene / proto-oncogene ACCEPT not functioning properly apoptosis inhibited metastasis secondary tumour forming	(2)

Question number	Answer	Additional guidance	Mark
2(b)(i)	An answer that includes three of the following points with at least one similarity and one difference: Similarities: • highest cause of death is lung and bronchus in both (1) • lowest cause of death is pancreas in both (1) • lung and bronchus cause the same number of deaths in both (1) Differences: • deaths from all cancers, except lung and bronchus, are higher in ethnic group 2 (1) • more people died of cancer in group 2 (than group 1) (1)	DO NOT PIECE TOGETHER NB responses must refer to {deaths / data} not height of bars ACCEPT similar ACCEPT {prostate / colon and rectum / breast / pancreas} are higher in group 2 converse for group 1 ACCEPT converse for group 1	(3)

Question number	Answer	Mark
2(b)(ii)	The only correct answer is D <i>A is incorrect as statement 1 is wrong as $(47-18) \div ((47+18) \div 2) \times 100 = 89.23$</i> <i>B is incorrect as statement 1 is wrong as $(47-18) \div ((47+18) \div 2) \times 100 = 89.23$</i> <i>C is incorrect as statement 1 is wrong as $(47-18) \div ((47+18) \div 2) \times 100 = 89.23$</i>	(1)

Question number	Answer	Additional guidance	Mark
2(b)(iii)	20 per hundred thousand / 2 x / 2.1 x / twice as many (1)	IGNORE 2 / 2.1	(1)

Question number	Answer	Additional guidance	Mark
2(b)(iv)	An answer that includes the following points: • (not very useful) as cancers cause different numbers of deaths in different (ethnic) groups (1) • (useful) if you are not in one of these (ethnic) groups as you get an idea of the effects of the different cancers (overall) (1)	ACCEPT other factors / named factors involved ACCEPT useful (to professionals) to focus on {highest causes of cancer / screening programmes / health initiatives}	(2)